

Kevin Cook

Hardware Architect, Aingura IIoT



PERSONAL DATA

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SUMMARY

Bilingual hardware architect, circuit designer, and project manager with a unique multi-sector background and extensive industry experience, including product development work in the industrial machine learning, satellite television and datacom sectors. Expertise in 10Gb circuit design for networking systems, PCB design for manufacturability, Bayesian networks, and development of software for design automation and product validation. Proven ability to collaborate with clients, partners, vendors, and contract manufacturers.

TECHNICAL EXPERTISE

Methodologies/Technologies

- Hardware-centered product development from initial concept, to requirements negotiation with clients, to functional prototype, through validation to mass production
- Circuit design, schematic capture, and PCB routing with Altium, Zuken, and KiCAD
- Project management including coordination of mechanical, hardware, software, web, and cloud developers
- Statistical modeling with Bayesian networks, visualization, and inference using the Python/NumPy/Jupyter stack applied to industrial data, circuit simulation, design automation, and product validation results
- Automated test development using HTML5+JavaScript, Python+Robot Framework, Ruby, Lua, TCP/IP, and bespoke fixtures
- Product validation and characterization using extensive automation, RF techniques, EMC anechoic chambers, BERT, time domain reflectometry, power, thermal, and vibration testing
- Development of HTML5+JavaScript web applications for internal R&D use

PROFICIENCIES

PROGRAMMING	Python+NumPy, R, HTML5+JavaScript, C/C++, Ruby+Cucumber, Java, Markdown
ELECTRONICS DESIGN	Altium, KiCAD, Zuken (CADSTAR, SI Verify), Mentor Graphics (DxDesigner, PADS Router), LTspice, Hyperlynx, Polar
ENGINEERING	FreeCAD, SketchUp, MATLAB/Octave/Simulink, Excel
VIDEO & NETWORKING	Corporate certification in HEVC, HLS/MPEG-DASH, G.984 GPON; Telcordia certification in TCP/IP, DWDM, SONET/SDH, T1/E1
SYSTEMS & PLATFORMS	Linux, Windows, Docker, VMware, Vagrant, EC2
LANGUAGES	English (native), Spanish (fluent)

PROFESSIONAL EXPERIENCE

Current	Hardware Architect at AINGURA IIoT (INZU Group) <i>Donostia-San Sebastián, Spain</i> Developing the next generation of hardware platforms for industrial sensing, edge computing, OT networking: Aingura Impulse industrial distributed data acquisition and machine learning system Responsible for hardware development for an integrated product line of modular DAQ and processing elements. Leading next-generation computing platform selection including benchmarking of CPU/memory subsystems. Aingura IPNI edge computing node for infrastructure Redesigned a modular compute unit for outdoor infrastructure. Collaborated with assembly providers and firmware developers for unsupervised remote operation. Led new chassis concept development. Aingura LAPA series of DAQ interfaces Developed data acquisition electronics specifically for bridges and overpasses. Currently deployed in Massachusetts and Spain.
AUG 2020	
JUL 2020	R&D Project Manager at IKUSI MULTIMEDIA (now part of Fagor Electrónica) <i>Donostia-San Sebastián, Spain</i> Managed project to develop customized, cloud-connected television headend Directed multi-year hardware/software/cloud project for a key customer, coordinating product, features, timelines, and validation. Platform is currently deployed in hundreds of sites across Australia. Led hardware and mechanical development for new multimedia headend Developed modular headend architecture including a novel, cost-effective networking system for control based on embedded USB—several current products are based on this platform and its derivatives. Implemented new active thermal system for the platform. Led hardware design of all key PCBAs for the system, integrating new microprocessor, FPGA, DRAM systems, and Ethernet IPcores. Collaborated with PCB fabricators to optimize performance and cost of hardware, with firmware developers to rationalize and modularize devicetree, and mechanical designers to improve EMC. Directed design of networking subsystem for TV platform Specified, designed, and implemented streaming-optimized networking system, from hardware to VLAN level, from line card PHYs to backplane to switch fabric ASIC. System based on gigabit Ethernet integrates switching ASIC in custom modular hardware, and forms the backbone of Ikusi's current premium TV headend. Switch module also includes microprocessor and FPGA for headend control. Managed development of custom firmware for switch ASIC by external contractor. Developed validation protocols, and production test stand including bespoke hardware and software written in Ruby.
MAY 2012	
MAY 2012	R&D Technical Staff at IKUSI MULTIMEDIA (now part of Fagor Electrónica) <i>Donostia-San Sebastián, Spain</i> Designed hardware for high-density H.264 transcoding module System based on ASIC with fast multi-channel DDR3 and a PCI-e interface. Characterized and selected ICs for embedded Ethernet subsystem. Simulated signal integrity of PCI-e lanes using Keysight ADS. Miniaturized PCB layout by a factor of two compared to the ASIC reference design. Benchmarked transcoding performance, optimized thermal management system, developed and executed validation protocols, and cost-reduced the design for mass production. Transcoding systems installed in thousands of B2B locations for a major television operator in France. Developed hardware for network-enabled satellite television transmodulator Module was first in its class to incorporate Ethernet networking. Simulated signal integrity of system bus using Zuken SI Verify. Improved DFM and cost of design by increasing integration, migrating PCB fabrication to high-volume overseas facilities, and optimizing mechanical design for assembly by robot. Developed and executed validation protocols, assisted production team with jigs and procedures for automation, and adapted hardware platform to new form factors for new products.
DEC 2007	

Nov 2007	Electrical Engineer at MOLEX INC. HIGH PERFORMANCE ADAPTERS <i>Maumelle, Arkansas, USA</i> Redesigned 4.25 gigabit/second Fibre Channel active copper cable assembly New design had updated transmitter IC to reduce EM emissions, exhibited improved manufacturability and durability. Collaborated with contract manufacturer and PCB fabricator to tune characteristic impedance to optimize signal integrity. Characterized the new assembly, which had improved signal quality, bit error rate, and lower overall cost. Developed 4.25 Gbps production test system including FPGA-based pattern generator Tester required four new printed circuit boards assemblies, three with tightly controlled impedance. Incorporated a customized Xilinx evaluation board programmed to generate test pattern. Coordinated development with mechanical engineer developing the enclosure, performed final assembly and testing, and shipped the tester to Mexico, where it was in heavy use in a production line.
JAN 2005	
SEP 2003	Member of Technical Staff at TELLABS OPERATIONS, INC. <i>Naperville, Illinois, USA</i> Designed gigabit passive backplane for optical telecommunications system Hardware deliverable was production quality from the first prototype, saving tens of thousands of dollars and months of delay by eliminating the need for a second prototype. Design was a key component in product with hundreds of millions of dollars of revenue per year. Developed automated test system in LabWindows/CVI Custom software-controlled instruments to confirm compliance with Telcordia T1/E1 standards. Verified 28 T1 ports with no human intervention. Designed gigabit passive backplane for optical telecommunications system Measured differential and single-ended signal paths, correlated test data to RF simulations, and performed 2.4 gigabit/second bit error rate testing, thoroughly documenting results.
JAN 2000	
MAY 1999	Undergraduate Research Assistant at IOWA STATE UNIVERSITY <i>Ames, Iowa, USA – Physics Department</i> Debugged hardware and software for custom VME-based research instrument Performed low-level testing, including diagnostic routines in C for the VxWorks platform, fault isolation, and device programming for the electronics which became the Level I Trigger for the PHENIX experiment at the Relativistic Heavy Ion Collider at Brookhaven National Laboratory.
JAN 1998	

EDUCATION

JULY 2026	University of the Basque Country (UPV/EHU) – Donostia-San Sebastián, Spain M.Eng. in COMPUTATIONAL ENGINEERING AND INTELLIGENT SYSTEMS (KISA) Master's Thesis: Bayesian Network Performance on Embedded Hardware: Optimizing Platform Selection for Energy Efficiency in Industrial Applications Machine Learning Coursework: Advanced Probabilistic Modeling, vision and convolutional neural networks, time series, bioinformatics, optimization, deep learning.
2018	Foundations of Data Science Professional Certificate (BerkeleyX, 2018)
DEC 2004	University of Illinois at Chicago – Chicago, IL, USA Specialized Physics Coursework: Quantum mechanics, thermodynamics, statistical mechanics, and electromagnetism. Research Assistantship: Volunteered in a thin-film magnetism laboratory; designed and assembled a cryogenic deposition platform.
AUG 2003	
DEC 1999	Iowa State University – Ames, IA, USA B.S. in ELECTRICAL ENGINEERING Communications Concentration: Digital communication systems, feedback control theory, and network programming (C, Java). Senior Project: JALTISA (Java Linear Time-Invariant System Analyzer)
AUG 1994	